

PENGFEI HU

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Education

Stevens Institute of Technology

Sep. 2023 – Present

Ph.D. in Computer Science (GPA: 3.96/4)

Hoboken, NJ

- **Ph.D. Advisor:** Dr. Yue Ning;
- **Research Interests:** Predictive Healthcare, Domain Generalization, Large Language Models, Graph Neural Network;
- Teaching Assistant for CS-584: Natural Language Processing on Fall 2024 and Spring 2025.

University of Southern California

Sep. 2021 – May 2023

M.S. in Industrial System Engineering (GPA: 4/4)

Los Angeles, CA

- Research Assistant advised by Dr. Sze-chuan Suen working on monitoring of diabetes via wearable sensor devices.

Nanjing University of Finance and Economics

Sep. 2016 – May 2020

B.A. in Economic Statistics (Rank: 1 / 145, GPA: 3.63/4)

Nanjing, China

- Working on relative deprivation in the digital era (1 publication) supervised by Dr. Qinghai Li.

Work Experience

Oak Ridge National Laboratory

May 2025 – Aug. 2025

Graduate Research Intern, Mentor: Ming Fan

Oak Ridge, TN

- Utilized the Frontier supercomputer server to deploy parallel training on over 10 compute nodes via DDP and mpi4py;
- **Project:** Provide multi-reservoir inflow predictive support for water management in the Upper Colorado River Basin;
- **Paper:** Two papers (1st author) will be submitted to workshops and further refined as journal submissions.

Alibaba Group, Inc.

Mar. 2021 – Jul. 2021

Data Analyst Intern

Hangzhou, China

- Tracked 20GB delivery data via Hive SQL to evaluate performance like freight volume and penetration of shipment to filter potential counties, calculated ROI and compared confident intervals to allocate ads budget efficiently by 10%;
- Developed 5 hourly or daily dashboards by business intelligence tools to regulate fluctuation of metrics (Budget, active/new users & couriers etc.);
- Summarized high-risk users and matched them with behaviour-based features (fake Mac-Address, pageview status, online duration etc.), constructed and tuned best model LightGBM by Python for prediction with 87% precision rate and saved 7% budget for fraud detection of risk control comparing with \$100k+ cost of last quarter.

Other Internships

- Product Analyst Intern at Tencent (Apr. 2020 – Aug. 2020, *Shenzhen, China*);
- Risk Data Analyst Intern at Didi Global (Dec. 2019 – Apr. 2020, *Chengdu, China*);
- Actuarial Analyst Intern at PwC Consulting (Jan. 2019 – Mar. 2019, *Shenzhen, China*).

Publications

(The google scholar link can be found here, * stands for equal contribution.)

1. **Pengfei Hu**, Chang Lu, Fei Wang, Yue Ning, “Bridging Stepwise Lab-Informed Pretraining and Knowledge-Guided Learning for Diagnostic Reasoning”, Under Review, 2024.
2. **Pengfei Hu**, Xiaoxue Han, Fei Wang, Yue Ning, “UdonCare: Hierarchy Pruning for Unseen Domain Discovery in Predictive Healthcare”, Under Review, 2025.
3. Xiaoxue Han, **Pengfei Hu**, Jun-En Ding, Chang Lu, Feng Liu, Yue Ning, “Interpretable and Interactable Predictive Healthcare with Knowledge-Enhanced Agentic Causal Discovery”, Accepted by EMNLP Findings, 2025.
4. Eric Yang*, **Pengfei Hu***, Xiaoxue Han, Yue Ning, “MPLite: Multi-Aspect Pretraining for Mining Clinical Health Records”, 2024 IEEE International Conference on Big Data (BigData), 2024.
5. Hsin-Ling Hsu, Cong-Tinh Dao, ..., **Pengfei Hu**, ..., Chenwei Wu (8th/15 authors in total), “MEDPLAN: A Two-Stage RAG-Based System for Personalized Medical Plan Generation”, Accepted by ACL Industry Paper, 2025.

Technical Skills

Programming Languages: Python, SQL, Java, R, Latex

Deep Learning / Machine Learning Framework: Pytorch, PyG, Tensorflow, scikit-learn